Reproducibility

Reproducibility I

The bundled `manuscript/config.yaml` is configured for the **strict reproducibility** discipline advocated for computational science by [@peng2011reproducible]:

1. **No network calls.** All analysis is local: prose metrics, structure, quality flags, and bibliography validation are computed from in-repo files only.
2. **Deterministic outputs.** `compute_metrics`, `analyze_structure`, and `analyze_quality` are pure functions over their input strings. The same manuscript text + the same `config.yaml` produces byte-identical JSON artefacts (modulo timestamp metadata in any caches the project later adds).
3. **Threshold transparency.** Every pass/fail decision is recorded in `output/checks.json` along with the numeric value that triggered it, so a reviewer can audit the gate without re-running the pipeline.

Reproducibility II

4. **No hidden state.** The pipeline does not mutate `manuscript/`; it reads and reports. `manuscript/references.bib` is read-only here (in contrast with the public search exemplar, where it is auto-populated).
5. **Same code, different config.** A reviewer who wants stricter standards edits `manuscript/config.yaml` (`prose.target_grade_level_max: 14.0`, say); no code changes are required.

Verifying reproducibility locally I

```
# Run twice; check there are no diffs in the JSON artefacts
uv run python projects/templates/template_prose_project/scr
mv projects/templates/template_prose_project/output project
uv run python projects/templates/template_prose_project/scr
diff -ru \
    projects/templates/template_prose_project/output_first/
    projects/templates/template_prose_project/output/manusc
```

The diff should be empty. If it is not, the pipeline has acquired non-determinism — file an issue.

Determinism guarantees I

Stage	Deterministic?	Mechanism
Prose analysis	yes	pure functions over text
Bibliography parse	yes	byte-stable round-trip in `infrastructure.reference.citation`
Check evaluation	yes	pure comparisons
Markdown report assembly	yes	sorted file list, fixed output template
Figure generation	yes (within Matplotlib version)	fixed palette, no random subsampling
Manuscript variables	yes	derived from the deterministic JSON report